# Technical Design: Multi-Source Candidate Data Transformer

## 1. Pipeline Breakdown

The transformer is a modular, deterministic engine executed sequentially:

1. **Detect & Extract:** Identifies input types (.pdf, .docx, .csv, .json, URLs) and invokes specialized parsers (e.g., PyMuPDF for PDF, regex/standard libraries for JSON/CSV).
2. **Normalize:** Extracted fields pass through pure functions to standardize formats globally (e.g., lowercasing emails, formatting dates, stripping whitespace).
3. **Canonicalize:** Unstructured skills or locations are mapped to standard canonical dictionaries.
4. **Merge (Entity Resolution):** Profiles are aggregated into a single record based on a predefined or UI-configured priority hierarchy, resolving conflicts deterministically.
5. **Project:** A dynamic runtime configuration layer reshapes the schema (renaming, omitting, raising errors on missing data).
6. **Validate & Output:** The final dictionary is passed through a strict Pydantic model (`CandidateProfile`) ensuring shape correctness before emitting JSON.

## 2. Canonical Output Schema & Normalizations

The canonical record acts as the internal source of truth:

- **emails (string[] ):** Normalized to lowercase, stripped of spaces.
- **phones (string[] ):** Normalized strictly to **E.164 format** via the `phonenumbers` library.
- **location (object):** Country names parsed to **ISO-3166 alpha-2** via `pycountry`.
- **skills (object[] ):** Raw terms mapped to canonical terms (e.g., 'JS' ➔ 'JavaScript').
- **experience (object[] ):** Start/End dates normalized to **YYYY-MM** format.
- **provenance (object[] ):** Field-level traceability dictating source name and method.

## 3. Merge Policy & Conflict Resolution

The engine resolves conflicts deterministically via a **Source Priority List** (e.g., `Resume > LinkedIn > ATS JSON > Recruiter CSV`).

- When mapping a field (e.g., `primary_email`), the engine iterates down the priority list. The highest-ranked source possessing a non-null, valid value "wins".
- **Confidence Assignment:** Confidence scores are assigned statically based on extraction method reliability. Structured sources (ATS JSON, CSV) yield high confidence (0.95-1.0), whereas unstructured regex extraction (Resume PDF) yields slightly lower confidence (0.80-0.90) due to potential parsing artifacts.

## 4. Runtime Custom-Output Configuration

A decoupled **Projection Layer** sits between the merge engine and the final output. It accepts a JSON payload to dynamically reshape the internal canonical record at runtime. It supports:

- **Subsetting & Renaming:** Filtering keys and renaming them via a `from` path (e.g., mapping `emails[0]` to `primary_email`).
- **Missing Value Policies:** Handling gaps by returning `null`, silently omitting the key, or raising an explicit `ValueError` to halt the run, depending on the strictness required by downstream consumers.

# 5. Edge Cases & Handling

1. **Garbage Strings & Encoding:** Handled via strict UTF-8 decoding during the Extractor phase. Unparseable tokens are ignored; unknown entities fall back to `null` to prevent crashing.
2. **Missing Required Sources:** The engine requires at least one source but does not crash if secondary sources are empty. It degrades gracefully, outputting partial profiles.
3. **Mismatched Phone Formats:** E.164 requires a country code. If a recruiter CSV provides "555-1234" (no country code), the normalizer assumes a default locale (e.g., US) or flags the provenance confidence as lower.
4. **Duplicate Candidates Across Same Source:** Not explicitly handled in this V1 scope to prioritize field-level merging over global dataset deduplication due to time pressure.
5. **Complex Skill Synonyms:** Handled via a simple static dictionary mapping. Complex contextual embeddings (LLM-based canonicalization) are deliberately left out of the core pipeline to maintain 100% determinism.